

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: MLA03
COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO4: Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. (BL3,BL4,BL5)

Assessment Tool Weightage: 40%

Dr G. A. SENTHIL

QUESTIONS: 25

Total Marks: 50

Annexure A

Design and Develop Model

DURATION: 4 HRS (2 Sessions)

Sampling-Based Planning

1. Draw a flowchart illustrating the sampling-based planning process in Model-Based Reinforcement Learning. (5 Marks)

Flowchart: Sampling-Based Planning in Model-Based Reinforcement Learning

Matter for Flowchart

Start



Observe Current State s



Use Learned/Known Environment Model

- Transition model $P(s'|s,a)$
- Reward model $R(s,a)$



Sample an Action a

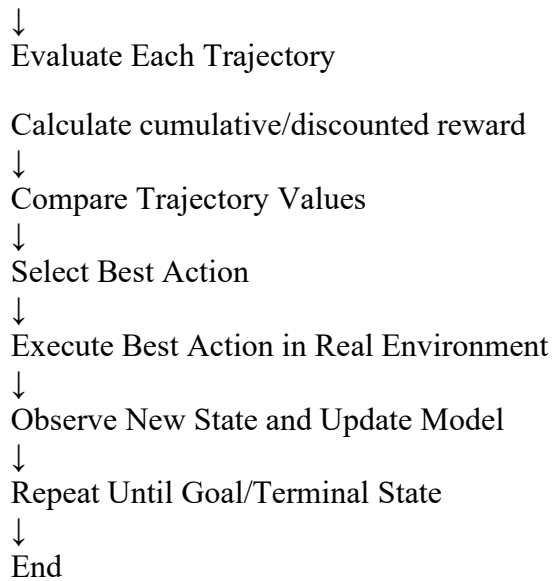


Predict Next State and Reward

$(s, a) \rightarrow (s', r)$



Generate Multiple Sample Trajectories



Key Point

Sampling-based planning avoids evaluating every possible state-action sequence. Instead, it samples promising trajectories from the learned model and selects the action with the highest expected return.

- 2. Prepare a concept map showing how sampled trajectories are generated and evaluated for planning. (5 Marks)**

Concept Map: Generation and Evaluation of Sampled Trajectories

Central Concept:

Sampling-Based Planning

Connect the following concepts:

Sampling-Based Planning

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1. Current State

- Starting state s_0
- Current environment information

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2. Environment Model

- Transition function
- Reward function
- Predicts future states

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3. Action Sampling

- Select possible actions
- Random or policy-based sampling
- Exploration of alternatives

↓

4. Trajectory Generation

$$\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \dots)$$

- Sample action
- Predict next state
- Receive/predict reward
- Continue for a fixed horizon

↓

5. Trajectory Evaluation

- Calculate total reward
- Calculate discounted return
- Estimate future value

Formula:

$$G = r_0 + \gamma r_1 + \gamma^2 r_2 + \dots$$

↓

6. Comparison

- Compare returns of trajectories
- Identify highest-value trajectory

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7. Action Selection

- Choose first action from the best trajectory
- Execute it in the environment

↓

8. Replanning

- Observe new state

- Generate new samples
- Repeat the process

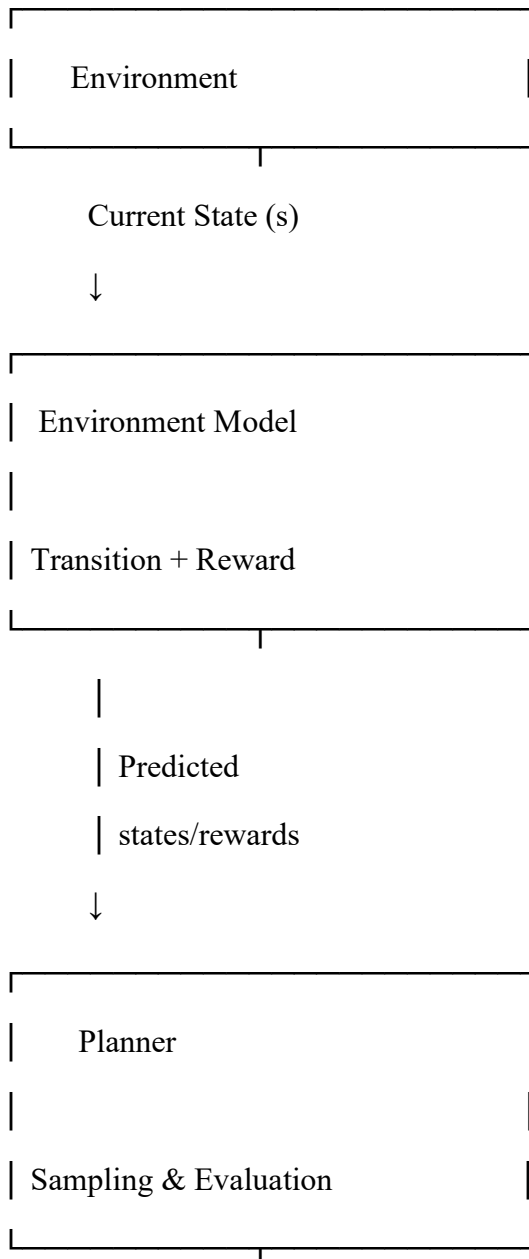
Overall Relationship

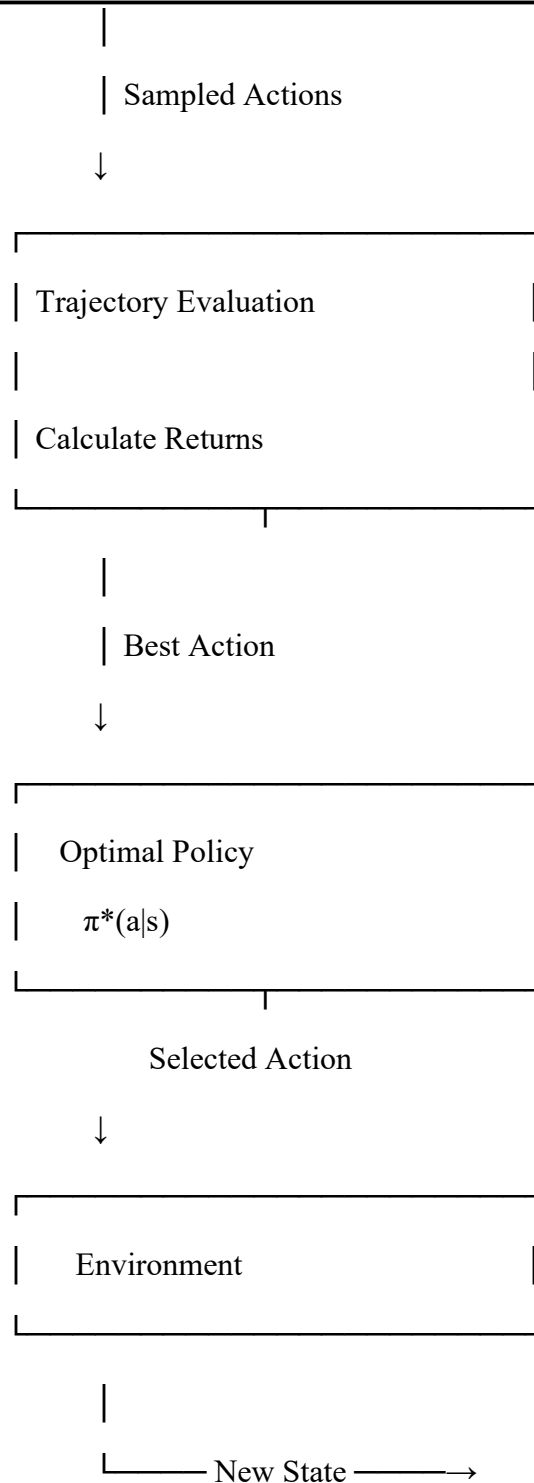
State → Model → Sample Actions → Generate Trajectories → Evaluate Returns → Compare → Select Best Action → Execute → New State → Repeat

3. Draw a block diagram showing the interaction between the environment model, planner, sampled actions, and optimal policy. (5 Marks)

Block Diagram: Environment Model, Planner, Sampled Actions and Optimal Policy

Draw the blocks in this order:





Block Descriptions

Environment: Provides the current state and receives the selected action.

Environment Model: Predicts the next state and reward for a given state-action pair.

Planner: Samples different actions and future trajectories using the model.

Trajectory Evaluation: Calculates the expected cumulative reward for each sampled trajectory.

Optimal Policy: Selects the action that provides the highest expected return.

4. Design a flowchart explaining Monte Carlo sampling for planning in Reinforcement Learning.

(5 Marks)

Flowchart: Monte Carlo Sampling for Planning in Reinforcement Learning

Matter for Flowchart

Start



Initialize Current State s_0



Select an Action

- Randomly or using current policy



Execute/Simulate Action



Observe Next State and Reward



Continue Sampling Until Terminal State or Fixed Horizon



Generate Complete Episode/Trajectory



Calculate Return G

$$G = r_0 + \gamma r_1 + \gamma^2 r_2 + \dots$$



Store Return for Visited State-Action Pairs



Generate Multiple Episodes



Average Returns

$$Q(s,a) = \text{Average}(G)$$



Compare Action Values



Select Action with Highest Estimated Value



Update Policy



Repeat Sampling and Planning



End

Key Point

Monte Carlo planning estimates the value of actions by generating complete sampled episodes and averaging their observed or simulated returns. It does not require the value of the next state to be known beforehand.

5. **Draw a concept map comparing exhaustive search planning and sampling-based planning techniques.**

(5 Marks)

Concept Map: Exhaustive Search Planning vs Sampling-Based Planning

Central Concept:

Planning Techniques in Reinforcement Learning

Split into two branches:

A. Exhaustive Search Planning

Exhaustive Search



- Explores all possible actions
- Examines all possible trajectories
- Produces highly accurate results for small problems
- Computationally expensive
- Requires large memory
- Becomes impractical for large state/action spaces
- Suitable for small and deterministic environments

Example:

Searching every possible path in a small grid.

B. Sampling-Based Planning

Sampling-Based Planning



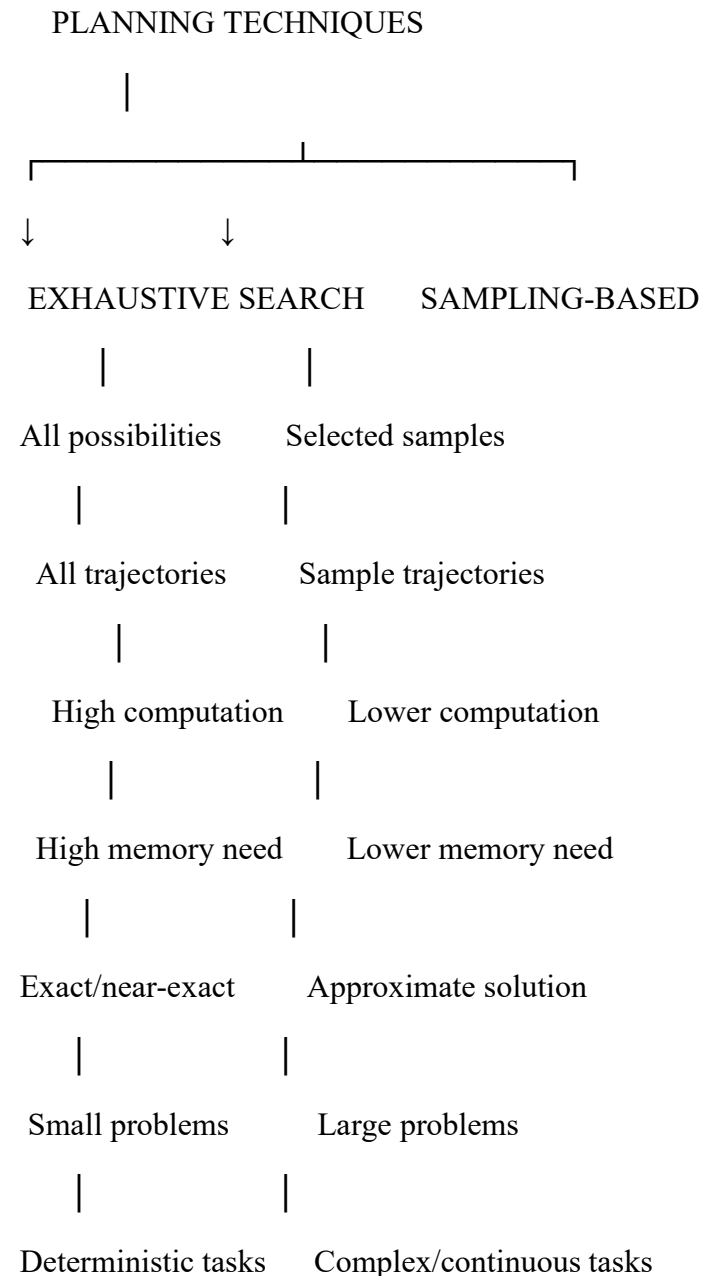
- Explores selected/sampled actions
- Generates a limited number of trajectories
- Uses environment model
- Evaluates sampled returns
- Reduces computational cost
- Requires less memory

- Suitable for large or continuous state spaces
- Results are approximate
- Can balance exploration and computation

Example:

Sampling several possible paths for a robot and selecting the path with the highest expected reward.

Comparison Map



Short Comparison Table

Feature	Exhaustive Search	Sampling-Based Planning
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Feature	Exhaustive Search	Sampling-Based Planning
Search space	Entire space	Sampled portion
Trajectories	All possible	Selected trajectories
Computation	Very high	Lower
Memory	High	Relatively low
Accuracy	High	Approximate
Scalability	Poor for large problems	Better
Suitable for	Small problems	Large/complex problems
Main advantage	Complete search	Efficient planning

Conclusion: Sampling-based planning provides a practical alternative to exhaustive search by evaluating only a subset of possible trajectories. This makes it particularly useful in large, complex, or continuous environments where evaluating every possibility is computationally expensive.

Code of Conduct:

I P. Sahith Sai (192425029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Sahith Sai

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness